

1(a)	F_v : kg m s^{-2} k : $\text{kg m s}^{-2} / \text{m} \times \text{m s}^{-1}$	C1
	$= \text{kg m}^{-1} \text{s}^{-1}$	A1
1(b)	$F = \rho g V$ $V = 4/3 \times \pi \times (2.1 \times 10^{-3})^3$ ($= 3.88 \times 10^{-8} \text{ m}^3$)	C1
	$\rho = 4.8 \times 10^{-4} / 9.81 \times V$	C1
	$= 1300 \text{ kg m}^{-3}$	A1
1(c)(i)	W downwards, U upwards, F_v upwards	B1
1(c)(ii)	$F_v = 7.2 \times 10^{-4} - 4.8 \times 10^{-4}$ $= 2.4 \times 10^{-4} \text{ (N)}$	C1
	velocity $= 2.4 \times 10^{-4} / (17 \times 2.1 \times 10^{-3})$ $= 6.7 \times 10^{-3} \text{ m s}^{-1}$	A1

Question	Answer	Marks
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